

Build & Ship with AI — Installation Guide

Codex · GitHub CLI · GitHub MCP · Chrome extension — a step-by-step setup handout for the 1-hour workshop

Goal of the workshop: use **Codex** (your AI coding assistant) plus **GitHub** to create a real project on your computer and publish it to GitHub — and be able to *explain and verify* everything that shipped.

Legend: WINDOWS Windows-only step macOS macOS-only step OPTIONAL advanced / terminal track

What you will install

Tool	What it is	Do you need it?
ChatGPT desktop app (Codex lives inside it)	Your main, point-and-click AI workspace for building projects.	Required
GitHub CLI (gh)	Publishes your project to GitHub with clear, repeatable commands.	Required
Git	The version-control engine gh and Codex rely on to save history.	Required
Codex Chrome extension	Lets Codex use a signed-in Chrome tab as context.	Recommended
Codex CLI (codex)	A terminal-first way to run Codex from a project folder.	Optional (advanced)
GitHub MCP server	Gives Codex a governed toolbelt to read/act on GitHub.	Optional (advanced)

Fastest path for a 1-hour session: do **Part 0** → **Part 4** (accounts, desktop app, GitHub CLI, Git, Chrome extension). Parts 5–6 (Codex CLI / GitHub MCP) are the advanced terminal track — set them up after the workshop if you run short on time.

Part 0 — Before you start: accounts & base checks

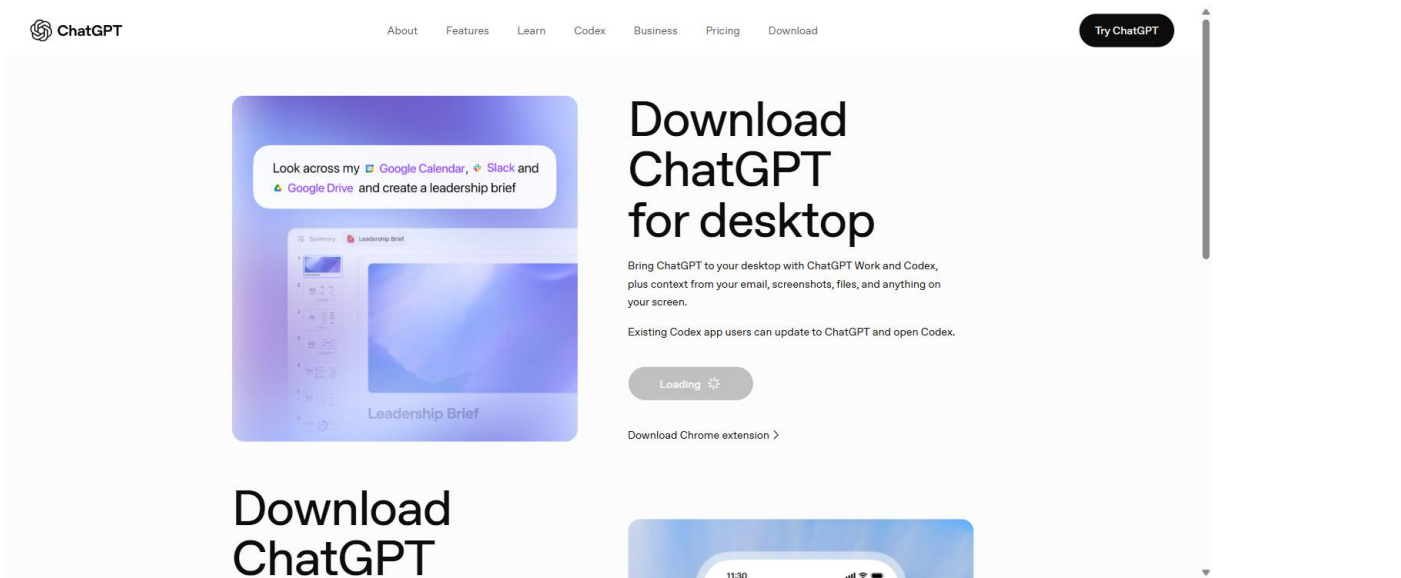
1. **GitHub account** with a **verified email**. Create one at github.com/signup. Students: apply for the free **GitHub Student Developer Pack** at education.github.com.

2. **ChatGPT account** that has access to **Codex**. Sign in at chatgpt.com and confirm you can see Codex.
3. **Google Chrome** installed (needed for the Chrome extension in Part 4). Get it at google.com/chrome.
4. **A quiet folder for the workshop**, e.g. `Documents\workshop` — you'll create your project here.

Credential rule (read this now): never paste a password or token into source code, a README, a prompt, or a committed `.env` file. Prefer browser sign-in, or store tokens in environment variables. If a real secret is ever exposed, **revoke/rotate it** — deleting the file is not enough because Git keeps history.

Part 1 — Install the ChatGPT desktop app & open Codex WINDOWS macOS

1. Go to the official download page: chatgpt.com/download and get the app for your operating system.
2. Run the installer, then **open the app** and **sign in** with the ChatGPT account from Part 0.
3. In the app, open the **Codex** workspace (from the model/workspace selector) and **start a new task**.
4. **Verify:** you can select a local folder and send a simple request (e.g. "list the files in this folder").



The official download page at chatgpt.com/download — note "Existing Codex app users can update to ChatGPT and open Codex," and the **Codex** item in the top nav.

SCREENSHOT TO ADD

The **Codex workspace** open inside the ChatGPT desktop app, with a local folder selected.

Capture during a real run — desktop-app UI can't be screenshotted from the web.

The product you download is named **ChatGPT**. **Codex** is a workspace *inside* that app — there is no separate "Codex desktop" download.

Part 2 — Install GitHub CLI (gh) & sign in [WINDOWS](#) [macOS](#)

Windows — open **PowerShell** and run:

```
winget install --id GitHub.cli --source winget
```

macOS — with [Homebrew](#) installed, run:

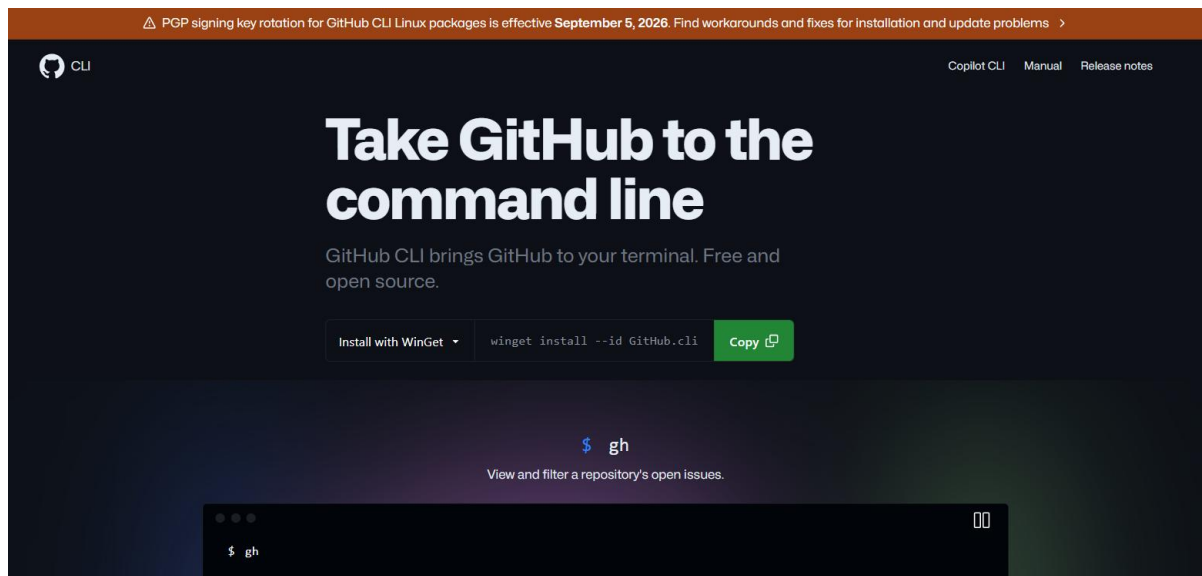
```
brew install gh
```

Then **close and reopen** the terminal and authenticate:

```
gh --version  
gh auth login
```

Choose **GitHub.com** → **HTTPS** → **Login with a web browser**, then paste the one-time code. When done, confirm:

```
gh auth status
```



The GitHub CLI home at cli.github.com — the install command for your platform is shown right on the page (Windows shown here).

SCREENSHOT TO ADD

Your terminal after `gh auth login`, showing the **one-time code** and the browser device-activation prompt.

Capture from your own terminal during setup.

The browser login flow means you never type or paste a token — the safest option for a classroom.

Part 3 — Install Git [WINDOWS](#) [macOS](#)

Git is the engine behind commits and pushes. Check first — it may already be installed:

```
git --version
```

If it is missing:

Windows:

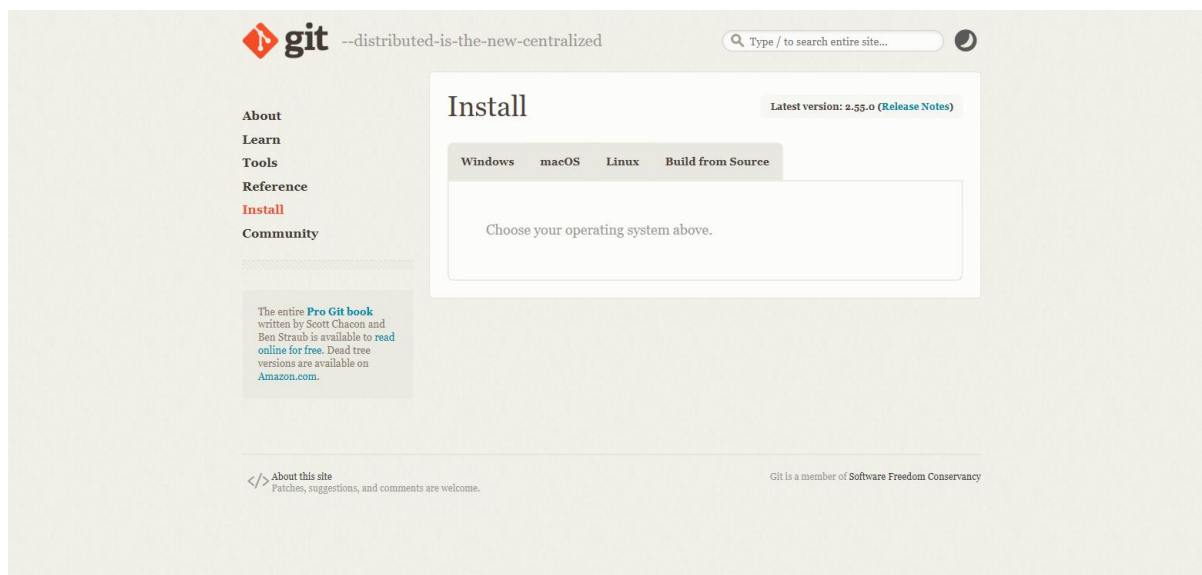
```
winget install --id Git.Git --source winget
```

macOS:

```
xcode-select --install # or: brew install git
```

Set your identity once (used on every commit):

```
git config --global user.name "Your Name"
git config --global user.email "you@example.com"
```



The Git install page at git-scm.com/downloads — pick your operating system tab, or just verify with `git --version` (it may already be installed).

Part 4 — Install & connect the Codex Chrome extension

The extension is installed **through the desktop app**, not straight from the Chrome Web Store.

1. Update and open the **ChatGPT desktop app**.
2. Open **Settings** → **Computer Use**.
3. Select **Chrome**, then click **Install** — it opens the extension's store page.
4. Install the extension and **approve the permission prompts**.
5. Return to **Computer Use** and confirm Chrome now shows **Manage** (that means it's connected).

SCREENSHOT TO ADD

The desktop app's **Settings** → **Computer Use** panel showing Chrome with a **Manage** button (confirms it's connected).

Capture from the desktop app after installing the extension.

Try it: start a Codex task, type `@Chrome` (or open side chat), and pick the Chrome profile where the extension is installed.

Safety: the extension can read and act on pages you're signed into. Treat webpage content as **untrusted instructions** — always review the target and the permission prompt before approving an action.

Don't confuse the connectors: the **Chrome extension** = signed-in *browser* context. **GitHub MCP** (Part 6) = structured *GitHub* tools. They do different jobs.

Part 5 — Install the Codex CLI **OPTIONAL**

The Codex CLI runs Codex from inside a project folder in your terminal. Pick **one** method.

macOS / Linux — native. Use npm (needs **Node.js 18+**) or the Homebrew cask:

```
npm install -g @openai/codex
# or
brew install --cask codex
```

Windows 11 — the Codex CLI runs inside **WSL2** (a real Linux environment on Windows), not PowerShell.

Open **PowerShell as Administrator** and install WSL + Ubuntu:

```
wsl --install
```

2. **Restart** your PC. Ubuntu finishes setup and asks you to create a Linux **username and password**.

Open the **Ubuntu** app (from Start menu). Install Node.js, then Codex, inside Ubuntu:

```
sudo apt update && sudo apt install -y curl git
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -
sudo apt install -y nodejs
npm install -g @openai/codex
```

Verify and launch (in the terminal where you installed it — Ubuntu on Windows):

```
codex --version
codex          # then choose "Sign in with ChatGPT"
```

If `codex` "is not found," close and reopen the terminal so the updated `PATH` loads.

Part 6 — Connect GitHub MCP to Codex **OPTIONAL**

GitHub MCP gives Codex a governed set of GitHub tools (repos, issues, pull requests). Easiest is the **remote** server hosted by GitHub.

1. Create a **fine-grained Personal Access Token (PAT)** at github.com/settings/personal-access-tokens/new with least-privilege scopes (start with `repo`; add `workflow`, `read:org` only if needed).

Put the token in an **environment variable** (never in code):

```
# macOS / Linux / WSL — add to ~/.bashrc or ~/.zshrc
export GITHUB_PAT_TOKEN="ghp_your_token_here"
```

3. Add the server to Codex — via the CLI:

```
codex mcp add github --url https://api.githubcopilot.com/mcp/ \
--bearer-token-env-var GITHUB_PAT_TOKEN
```

...or by editing `~/.codex/config.toml`:

```
[mcp_servers.github]
url = "https://api.githubcopilot.com/mcp/"
```

```
bearer_token_env_var = "GITHUB_PAT_TOKEN"
```

4. **Verify:** in the Codex TUI run `/mcp` and confirm **github** lists tools, then ask: "List my GitHub repositories."

Start read-only. Use MCP to *read* context first; enable write actions only when the task needs them, and always review a proposed action before approving it.

Part 7 — Move between conversations (the switches)

Keep **one conversation per project or feature** so context stays clean. Here's how to leave a chat and come back.

In the terminal (Codex CLI)

Command	What it does
<code>codex resume</code>	Opens a picker of recent chats — choose one to reopen.
<code>codex resume --last</code>	Jumps straight into your most recent chat (skips the picker).
<code>codex resume <id name></code>	Reopens one specific saved chat by its id or a name you gave it.

Inside a chat (type these in the Codex prompt)

Command	What it does
<code>/clear</code>	Clears the screen and starts a fresh chat.
<code>/rename</code>	Names the current chat so you can find it later with <code>resume</code> .
<code>/agent</code>	Switches the active agent thread.

In the ChatGPT desktop app

Each Codex **task** is **its own thread** in the history sidebar. Click a past task to return to it; start a **new task** for new work. Sessions are stored locally under `~/codex/sessions/`.

Habit to build: name a chat with `/rename` when you start real work, and reopen it with `codex resume` next time — instead of starting over and losing context.

Part 8 — Your first project: build with Codex, push with GitHub

Describe — in Codex, point at your workshop folder and give a clear prompt with a goal, constraints, and proof:

"Create a one-page personal site using HTML and CSS, plus a README with setup steps. Keep it accessible and responsive. **Before** changing files, tell me your plan. **After**, list every file you changed."

2. **Review** — read Codex's plan and the file diff before accepting. Open the page locally to confirm it works.

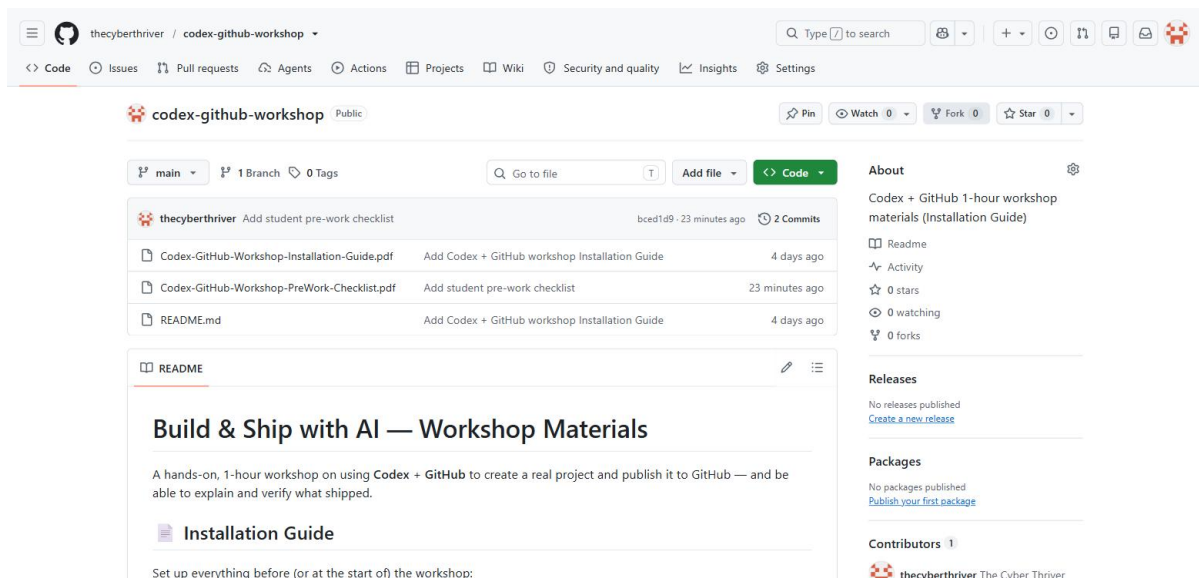
Publish — in your terminal, from the project folder:

```
gh auth login                # if not already signed in
git init
git add .
git commit -m "Initial project"
gh repo create --source=. --public --push
```

Inspect — confirm it's live:

```
gh repo view --web
```

4. Compare the GitHub page against your local folder — they should match.



What "done" looks like on GitHub — a public repo with your files and a README rendering on the home page (this is the workshop's own repo).

Definition of done: a public repository with a readable README, one reviewed commit, and a working project preview — and you can explain what each command did.

Part 9 — Final verification checklist

Check	How
ChatGPT desktop app opens Codex	App launches, Codex task opens, local folder visible
GitHub CLI works & signed in	<code>gh --version</code> and <code>gh auth status</code>
Git works & identity set	<code>git --version</code> , <code>git config --global user.name</code>
Chrome extension connected	Settings → Computer Use shows Manage for Chrome
Codex CLI (optional)	<code>codex --version</code> then <code>codex</code> signs in
GitHub MCP (optional)	<code>/mcp</code> lists github tools

Part 10 — Troubleshooting

Symptom	Fix
Command "not found" right after installing	Close and reopen the terminal so <code>PATH</code> refreshes.
<code>winget</code> not recognized (Windows)	Update App Installer from the Microsoft Store, then retry.
<code>wsl --install</code> asks for a reboot / does nothing	Restart the PC; on first Ubuntu launch set a Linux username & password.
<code>gh auth login</code> browser code fails	Re-run it; make sure you're signed into the right GitHub account in the browser.
MCP <code>github</code> server not listed	Check the table name is exactly <code>[mcp_servers.github]</code> and the token env var is set.

MCP 401 Unauthorized

Token expired or missing `repo` scope — create a new PAT and update the env var.

Appendix — Command quick reference

```
gh --version           # check GitHub CLI
gh auth login          # sign in (browser)
gh auth status         # confirm signed in
git init               # start tracking a folder
git add .              # stage all changes
git commit -m "message" # save a snapshot
gh repo create --source=. --public --push # create remote + push
gh repo view --web     # open the repo in a browser
codex --version        # check Codex CLI
codex                  # start Codex in this folder
codex resume --last    # reopen your last chat
```

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